

Rohan Shinde

+1-213-234-8020 | rohanshi@usc.edu | [linkedin.com/in/rohanshinde24](https://www.linkedin.com/in/rohanshinde24) | github.com/rohanshinde24 | [Portfolio](#) | Los Angeles, CA

EDUCATION

University of Southern California	Los Angeles, CA
Master of Science in Computer Science GPA: 3.71	Jan 2025 – Dec 2026 (Expected)
Coursework: Analysis of Algorithms, Information Retrieval, Applied NLP, Database Systems, ML for Data Science	
Maharashtra Institute of Technology	Pune, MH
Bachelor of Technology in Computer Science and Engineering GPA: 8.93	2020 – 2024
Coursework: Software Engineering, Distributed Systems, Operating Systems, Big Data Technologies	

EXPERIENCE

Amazon – SCOT, Fulfill to Promise	May 2026 – Aug 2026
Software Development Engineer Intern	Bellevue, WA

- Designed, tested, and deployed a production **Model Context Protocol (MCP)** server with four Java tools using **AWS Lambda, API Gateway, DynamoDB, and S3**, achieving over **95%** unit test coverage
- Re-engineered retrieval of relevant historical plans from thousands of time-stamped S3 objects per order, replacing paginated listings with DynamoDB date-range queries to **accelerate lookup and reduce agent-context usage**
- Integrated the MCP with an internal troubleshooting service, accelerating fulfillment-decision analysis for approximately **5,000 internal users** and reducing on-call investigation time by **70%**, from **50 to 15 minutes**
- Owned end-to-end delivery across multiple supply-chain teams, coordinating requirements with engineers, PMs, and BI analysts and driving production readiness through AWS **CDK** and **CloudWatch observability**
- Utilized AI productivity tools, including **Claude and Kiro**, to streamline system-design research, **spec-driven development**, and debugging while evaluating technical tradeoffs before incorporating suggestions into the service

University of Southern California – Advancement Services	May 2025 – Feb 2026
Software Engineer Intern	Los Angeles, CA

- Engineered a full-stack search system using **React** with **TypeScript** and SharePoint **REST APIs**, integrating NLP-based intent recognition and similarity scoring to optimize query precision and response efficiency
- Developed an automated data integration pipeline in **Python** connecting Tableau and SharePoint via REST APIs, implementing token-based authentication to serve **440+ users** across 19 departments
- Cut Cognos query time by **80%** on multi-million-row PostgreSQL datasets using set-based rewrites and indexing

DMI Finance	Jul 2024 – Oct 2024
Software Engineer Intern	Delhi, India

- Fine-tuned Llama 3.1 with **QLoRA** for 35-class transaction classification, improving precision by **2.9x** over baseline
- Built a hybrid **vLLM** and **Groq** data-labeling pipeline, reducing model iteration time by **65%**

PROJECTS

QueryLens (GitHub) <i>Java, Spring Boot, PostgreSQL, REST API, Docker, TDD, CI/CD</i>
- Built a PostgreSQL analyzer with REST APIs for slow-query diagnostics and execution-plan analysis - Automated strategy-based SQL anti-pattern detection and rewriting, reducing benchmark latency by over 80%
RetentionPulse (Demo) <i>React, TypeScript, FastAPI, Python, Docker, CI/CD, pytest</i>
- Built FastAPI microservices with SHAP explanations for telecom churn prediction, achieving 0.84 ROC-AUC - Implemented async routing, connection pooling, and batch inference with CI-gated unit and integration tests

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks & Testing: Spring Boot, React, Flask, FastAPI, Node.js, JUnit, pytest
Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, DynamoDB, CloudWatch, CDK, EC2, RDS), Docker, Kubernetes, Linux, Git
Databases & ML: PostgreSQL, Redis, PyTorch, TensorFlow, scikit-learn, Spark, Pandas, vLLM, RAG, SHAP, RL
Concepts: Data Structures & Algorithms, Distributed Systems, Multithreading, Information Retrieval, Agile